CLASS 8 SCIENCE

Advanced Cell Structure & Organelles

Exploring complex cellular machinery and primary cell types.

Cell Classification

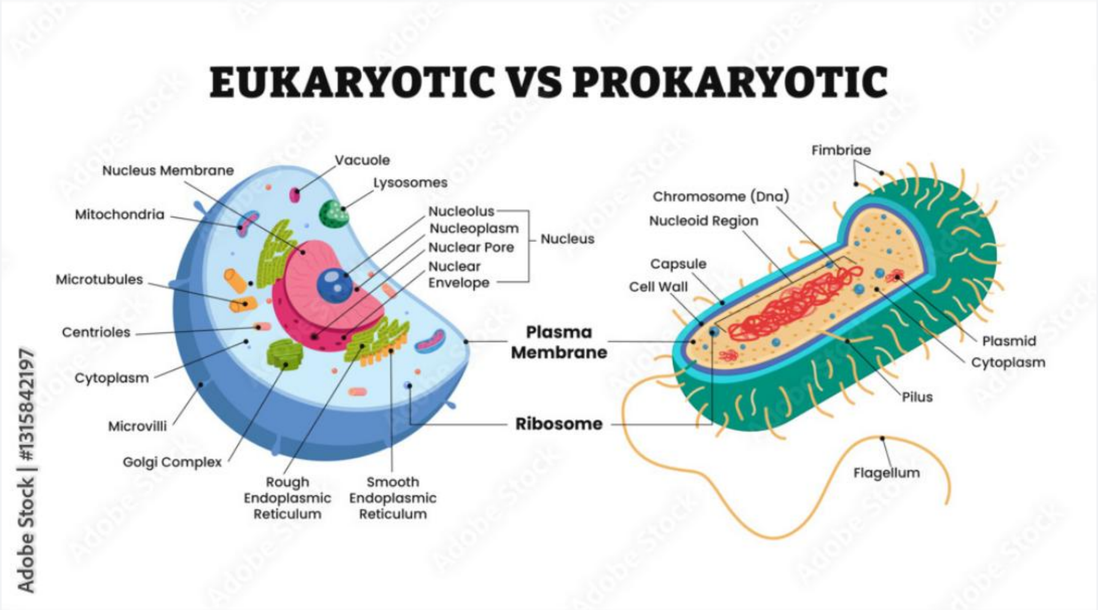
Prokaryotic Cells

- No true nucleus.
- Lacks membrane-bound organelles.
- Example: Bacteria.

Eukaryotic Cells

- Well-defined nucleus.
- Complex internal organelles.
- Example: Plants, Animals.

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ORGANELLES

Endoplasmic Reticulum (ER)

A vast network of membrane-bound tubes and sheets.

- **Rough ER:** Studded with ribosomes.
- **Smooth ER:** Lacks ribosomes entirely.
- **Function:** Synthesizes proteins and lipids.

ORGANELLES

Ribosomes

Protein Factories

Synthesizes proteins for the cell.

Appearance

Dense, granular spherical structures.

Location

Free-floating or attached to ER.

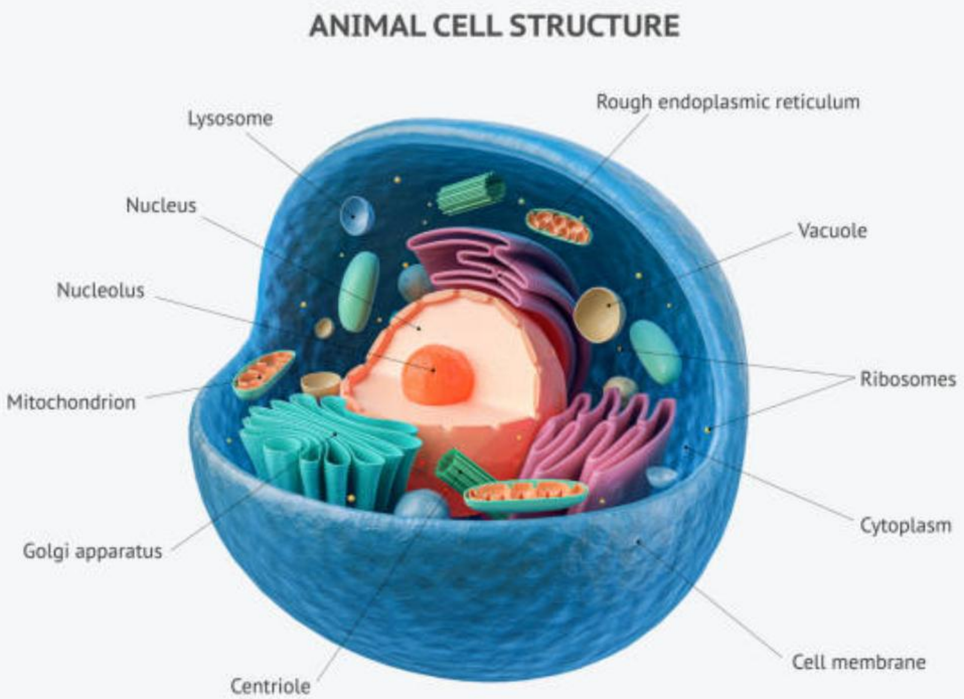
ORGANELLES

Golgi Bodies

Structure: Stacked, flattened membrane sacs.

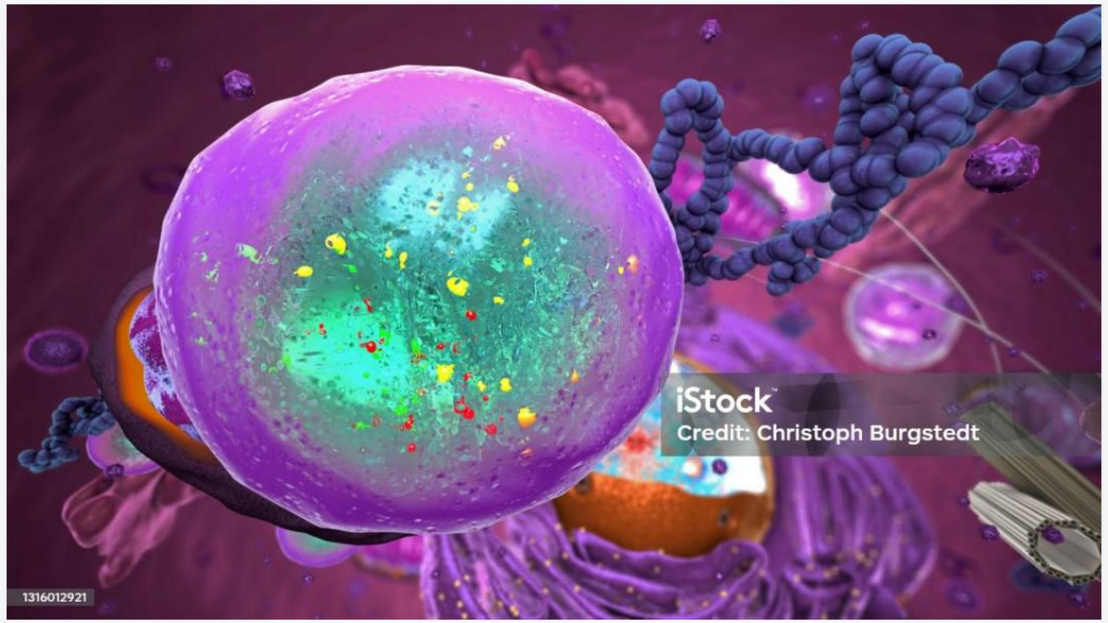
Function: Modifies, sorts, and packages.

Delivery: Dispatches materials via vesicles.



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ORGANELLES

Lysosomes

The waste disposal system of the cell.

- **Contents:** Powerful digestive enzymes.
- **Role:** Breaks down cellular waste.
- **Alias:** Often called "suicide bags".

PLANT CELLS ONLY

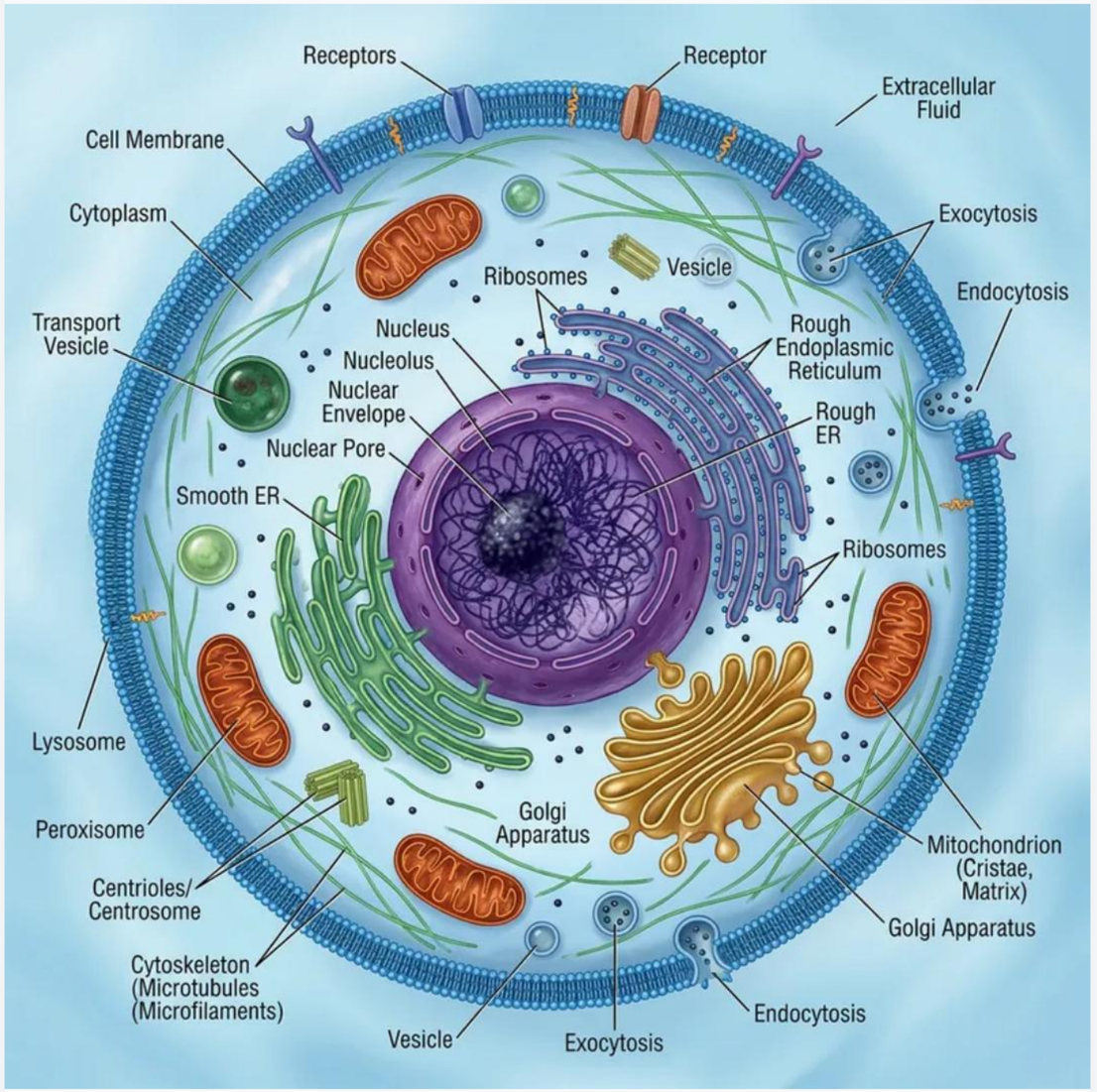
Plastids

- **Chloroplasts:** Green, handles photosynthesis.
- **Chromoplasts:** Contains colorful pigments.
- **Leucoplasts:** Colorless, stores starches.



Source: thejigsawpuzzles.com

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ANIMAL CELLS ONLY

Centrosomes

- **Location:** Found near the nucleus.
- **Structure:** Contains two centrioles.
- **Function:** Vital for animal cell division.



Assessment Time

One-Word Answers

1. Which organelles are commonly known as the "suicide bags" of the cell?

2. What type of cell lacks a well-defined nuclear membrane?

Short Answer Questions

Q1. State the primary function of ribosomes.

Q2. How does Rough ER differ from Smooth ER?

Q3. What is the specific role of chloroplasts in plant cells?

Q4. Why are centrosomes important in animal cells?

Q5. Briefly describe the function of the Golgi bodies.

Long Answer Questions

Question 1

Explain the structural and functional differences between prokaryotic and eukaryotic cells, providing one example of each.

Question 2

Describe the different types of plastids found in plant cells and state their specific functions.

Question 3

Discuss the interconnected roles of the Endoplasmic Reticulum and Golgi Bodies in cellular packaging and transport.

End of Topic

Prepare your answers for the next class session.

Image Sources



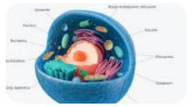
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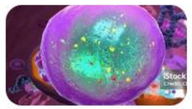
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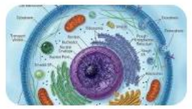
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